

Michael Pham

ktm-p.net

Email: ktmichael.pham@gmail.com

Mobile: (916)-968-0563

EDUCATION

- **University of California, Berkeley**

B.A. in Computer Science and Mathematics

Minor in Data Science

Berkeley, CA

Aug 2022 – Present

- GPA: 3.85
- Member of Upsilon Pi Epsilon Honor Society
- Dean's List, Honors to Date

PROFESSIONAL EXPERIENCE

- **Self-Employed**

Calculus Tutor

West Sacramento, CA

June 2025 – Present

- Private tutor for Calculus I and II.
- Created lessons tailored to individual students' needs, reinforcing key concepts to them.

PROJECTS

- **Machine Learning** | Python, PyTorch

- Used PyTorch on a variety of machine learning problems.
- Utilized a two-layer Recurrent Neural Network for language recognition. Achieved an accuracy of over 80%.
- Implemented a two-layer Linear Neural Network for handwriting recognition. Achieved an accuracy of over 98%.

- **Reinforcement Learning** | Python, PyTorch

- Utilized Reinforcement Learning to train Pac-Man agent to win. Achieved a win rate of over 90%.
- Implemented value iteration, Q-Learning, Approximate Q-Learning, and Deep Q-Learning using PyTorch.
- Used Multi-Layered Linear Neural Network in Deep Q-Learning.

- **Spam Classifier** | Python, Matplotlib, NumPy, Pandas, RegEx, scikit-learn, Seaborn

- Created a spam email filter using a Logistic Regression model. Achieved an accuracy of 99.2% on given test data.
- Cleaned and visualized data using Pandas, RegEx, Matplotlib, and Seaborn.
- Fine-tuned hyperparameters by cross-validation with GridSearchCV.

- **Image Classifier** | Python, NumPy, Pandas, scikit-learn, Seaborn, PyTorch

- Implemented a Multi-Layered Perceptron classifier for the Fashion MNIST data set.
- Added metrics and utilised visualisations and confusion matrix to analyse performance.
- Augmented images (rotations, flipping, scaling, etc.) to improve model's robustness.

- **Implementing the GGH Cryptosystem** | Python

- Wrote a paper discussing the mathematics behind the GGH Encryption Scheme and implemented it in Python.
- Implemented proper key generation, encryption, and decryption
- Minimized errors by switching to sympy. Optimized key generation to handle larger dimensions.

RELEVANT COURSEWORK

- **Computer Science:** Data Structures, Discrete Mathematics, Efficient Algorithms and Intractable Problems, Introduction to Artificial Intelligence, Introduction to Machine Learning, Optimization Models in Engineering
- **Mathematics:** Introduction to Analysis, Abstract Algebra, Abstract Linear Algebra, Numerical Analysis, Programming for Mathematical Applications
- **Data Science:** Principles and Techniques of Data Science, Probability for Data Science, Introduction to Ecological Data Analysis

TECHNICAL SKILLS

- **Programming Languages:** C, Golang, Java, MATLAB, Python, R, SQL
- **Frameworks/Libraries:** Matplotlib, Numpy, Pandas, Plotly, PyTorch, scikit-learn, Seaborn, TensorFlow
- **Tools:** Docker, gdb, git, Logism, LaTeX, Valgrind